

Chapter 9: Feature Detection & Feature Description

A Comprehensive Theoretical & Algorithmic Treatise on Classical and Deep Learning Local Descriptors

Local feature detection and description constitute the foundational backbone of computer vision tasks such as Structure-from-Motion (SfM), Visual Simultaneous Localization and Mapping (V-SLAM), multi-view stereo, and satellite photo-registration. This chapter provides an exhaustive theoretical analysis of both classical hand-crafted feature detectors/descriptors and modern deep learning-based local representation architectures.

Part 1: Classical Hand-Crafted Paradigms

ALGORITHM NAME: Harris Corner Detector

PARADIGM: Classical Differential / Auto-Correlation Matrix

THEORETICAL FOUNDATION: Evaluates intensity changes resulting from small local image shifts (u,v) using the auto-correlation matrix M . A corner is defined as a point where intensity varies significantly in all directions, corresponding to two large eigenvalues of M .

MATHEMATICAL FORMULATION:

$$M = \sum w(x,y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$
$$\text{Response } R = \det(M) - k \cdot (\text{trace}(M))^2 = (\lambda_1 \lambda_2) - k \cdot (\lambda_1 + \lambda_2)^2$$

ALGORITHMIC STEPS:

1. Compute image spatial gradients I_x and I_y using Sobel kernels.
2. Calculate gradient products I_x^2 , I_y^2 , and $I_x I_y$.
3. Convolve gradient products with a Gaussian window function $w(x,y)$ to construct matrix M .
4. Evaluate corner response function R using sensitivity constant $k \in [0.04, 0.06]$.
5. Apply non-maximum suppression (NMS) on R to isolate localized peak corners.

INVARIANCE PROPERTIES: Rotation-invariant; Intensity shift-invariant. **Not scale-invariant.**

COMPUTER VISION BENEFIT: Provides extremely fast and accurate 2D corner localization in stable, fixed-scale imaging scenarios.

ALGORITHM NAME: FAST (Features from Accelerated Segment Test)

PARADIGM: Classical Machine Learning / Direct Intensity Comparison

THEORETICAL FOUNDATION: Evaluates candidate pixel p by testing a Bresenham circle of 16 pixels surrounding p . A point p is classified as a corner if N contiguous pixels on the circle are all brighter or darker than I_p by a threshold t .

ALGORITHMIC STEPS:

1. Select candidate pixel p with intensity I_p .
2. Examine Bresenham circle pixels 1, 9, 5, and 13. If at least 3 do not exceed threshold criteria ($I > I_p + t$ or $I < I_p - t$), reject p immediately.
3. If 3 pixels pass, test all 16 perimeter pixels for a continuous sequence of N (typically $N=9$ or 12) passing pixels.
4. Optimize detection ordering using a ID3 decision tree compiled over training patches.
5. Apply score-based Non-Maximum Suppression (NMS) over adjacent candidate points.

INVARIANCE PROPERTIES: High computational throughput. **Lacks intrinsic scale and orientation invariance.**

COMPUTER VISION BENEFIT: Ideal for real-time edge devices and high-frequency Visual Odometry (VO) front-ends.

ALGORITHM NAME: SIFT (Scale-Invariant Feature Transform)

PARADIGM: Classical Scale-Space Gradient Histogram

THEORETICAL FOUNDATION: Detects extrema in Difference-of-Gaussians (DoG) space across octave scales, building a 128-dimensional spatial gradient orientation histogram descriptor around keypoints.

ALGORITHMIC STEPS:

1. Construct Gaussian scale-space octaves and compute Difference-of-Gaussian (DoG) images: $D(x,y,\sigma) = L(x,y,k\sigma) - L(x,y,\sigma)$.
2. Identify local extrema across $3 \times 3 \times 3$ spatial and scale neighborhoods.
3. Refine keypoint location to sub-pixel accuracy via 3D Taylor series expansion and eliminate low-contrast/edge points via Hessian trace/det ratios.
4. Assign dominant canonical orientation based on local gradient magnitude and direction histogram.
5. Construct $4 \times 4 \times 8$ spatial grid around keypoint, forming 8-bin orientation histograms per sub-region ($4 \times 4 \times 8 = 128$ dimensions), normalized to unit length.

INVARIANCE PROPERTIES: Fully scale, rotation, and illumination affine invariant; robust to mild perspective distortion.

COMPUTER VISION BENEFIT: The historical gold standard for geometric matching, visual mapping, and photogrammetric triangulation.

ALGORITHM NAME: SURF (Speeded-Up Robust Features)

PARADIGM: Classical Integral Image / Wavelet Approximations

THEORETICAL FOUNDATION: Approximates Gaussian second-order derivatives using box filters evaluated via integral images, accelerating keypoint detection and Haar wavelet descriptor computation.

ALGORITHMIC STEPS:

1. Construct Integral Image $I_{\Sigma}(x,y)$ enabling $O(1)$ box filtering.
2. Compute approximate Hessian matrix determinant using square box filters across scale spaces.
3. Locate scale-space maxima via 3D non-maximum suppression.
4. Determine dominant orientation using Haar wavelet responses in x and y within a circular neighborhood.
5. Construct a 64-dimensional descriptor vector by summing Haar wavelet responses ($\sum d_x, \sum d_y, \sum |d_x|, \sum |d_y|$) over 4×4 sub-regions.

INVARIANCE PROPERTIES: Scale and rotation invariant; significantly faster than SIFT.

COMPUTER VISION BENEFIT: High computational efficiency for real-time feature matching applications.

ALGORITHM NAME: ORB (Oriented FAST and Rotated BRIEF)

PARADIGM: Classical Binary Descriptor with Intensity Centroid Orientation

THEORETICAL FOUNDATION: Combines a scale-pyramid FAST detector with Intensity Centroid orientation steering and a learning-based decorrelated binary test pattern (rBRIEF).

ALGORITHMIC STEPS:

1. Detect FAST keypoints across an image scale pyramid.
2. Compute corner orientation using Intensity Centroid vector $C = [m_{10}/m_{00}, m_{01}/m_{00}]$, assigning angle $\theta = \text{atan2}(m_{01}, m_{10})$.
3. Steer 256 pre-selected binary test point pairs according to keypoint orientation θ .
4. Perform bitwise intensity comparison tests to yield a 256-bit (32-byte) binary descriptor string.

INVARIANCE PROPERTIES: Rotation and scale invariant; extremely fast matching via Hamming distance.

COMPUTER VISION BENEFIT: Essential for computationally constrained real-time systems like ORB-SLAM.

ALGORITHM NAME: BRISK (Binary Robust Invariant Scalable Keypoints)

PARADIGM: Classical Concentric Circular Sampling Binary Descriptor

THEORETICAL FOUNDATION: Uses a deterministic concentric pattern of sampling points to extract scale/rotation invariant binary descriptors evaluated via Gaussian smoothing.

ALGORITHMIC STEPS:

1. Detect keypoints in continuous scale space using AGAST (Adaptive and Generic Corner Detection).
2. Sample point intensities across concentric circles centered on the keypoint.
3. Compute global orientation vector $\mathbf{g}$ using long-distance point pairs.
4. Rotate sampling pattern by $\mathbf{g}$ and perform short-distance pairwise intensity comparisons to yield a 512-bit binary descriptor.

INVARIANCE PROPERTIES: Scale and rotation invariant; highly efficient bitwise distance evaluation.

COMPUTER VISION BENEFIT: High speed matching performance with higher precision than raw BRIEF.

ALGORITHM NAME: AKAZE (Accelerated-KAZE)

PARADIGM: Classical Non-Linear Diffusion Scale-Space

THEORETICAL FOUNDATION: Replaces Gaussian blurring with Fast Explicit Diffusion (FED) in non-linear scale space to smooth image noise while preserving sharp object boundaries and structural edges.

ALGORITHMIC STEPS:

1. Construct non-linear scale space using Fast Explicit Diffusion (FED) partial differential equations.
2. Compute Hessian determinant response across non-linear scale levels.
3. Extract keypoints using 3D non-maximum suppression.
4. Extract M-LDB (Modified Local Difference Binary) descriptors capturing gradient and intensity sub-grid dynamics.

INVARIANCE PROPERTIES: Scale, rotation, and illumination invariant; superior edge preservation.

COMPUTER VISION BENEFIT: High keypoint repeatability on complex natural textures, boundaries, and low-contrast regions.

Part 2: Deep Learning Neural Paradigms

ALGORITHM NAME: SuperPoint

PARADIGM: Deep Self-Supervised Joint Detector & Descriptor Architecture

THEORETICAL FOUNDATION: A fully convolutional neural network trained via Homographic Adaptation that jointly extracts interest point locations and dense semi-dense descriptors in a single forward pass.

ALGORITHMIC STEPS:

1. Pass $H \times W$ image through a shared VGG-style encoder to extract latent features.
2. Detector Head processes features into $H/8 \times W/8 \times 65$ tensor, yielding sub-pixel keypoint logits via Softmax and Depth-to-Space reshaping.
3. Descriptor Head computes dense $H/8 \times W/8 \times 256$ vector maps, bicubically interpolated and L_2 -normalized to 256D keypoint descriptors.
4. Train using synthetic shapes pre-training followed by real-image self-supervision via homographic warp aggregation.

INVARIANCE PROPERTIES: Robust to severe illumination changes, non-linear noise, and dynamic range shifts.

COMPUTER VISION BENEFIT: Modern benchmark for deep visual odometry and robust visual localization pipelines.

ALGORITHM NAME: D2-Net (Detect-and-Describe Network)

PARADIGM: Deep Dense Feature Map Joint Extraction

THEORETICAL FOUNDATION: Postpones detection decisions until deep layers, defining keypoints as local maxima in spatial and channel dimensions of dense convolutional feature maps.

ALGORITHMIC STEPS:

1. Pass image through CNN backbones (e.g., VGG-16 or ResNet) to generate dense feature tensor $\mathbf{F} \in \mathbb{R}^{h \times w \times C}$.
2. Compute hard spatial detection map a_{ij}^k and soft channel detection map b_{ij}^k .
3. Multiply spatial and channel selection maps to extract unified score map $SS_{ij} = \max_k (a_{ij}^k \cdot b_{ij}^k)$.
4. Extract keypoint locations at local maxima of SS_{ij} and sample corresponding L_2 -normalized sub-vectors from $\mathbf{F}$ as descriptors.

INVARIANCE PROPERTIES: Highly resilient to extreme illumination and multi-seasonal environmental variations.

COMPUTER VISION BENEFIT: Unmatched localization reliability under difficult lighting and extreme domain changes.

ALGORITHM NAME: R2D2 (Repeatable and Reliable Detector and Descriptor)

PARADIGM: Deep Reliability and Repeatability Prediction Network

THEORETICAL FOUNDATION: Jointly predicts keypoint locations, descriptors, local repeatability, and matching reliability to avoid unmatchable keypoints (such as sky or repetitive textures).

ALGORITHMIC STEPS:

1. Extract dense feature maps using L2-Net style fully convolutional backbone.
2. Predict dense 128D descriptors along with two explicit scalar maps: Repeatability R and Reliability S .
3. Evaluate descriptor reliability using a local average AP (Average Precision) loss over local neighborhoods.
4. Multiply repeatability and reliability maps ($R \cdot S$) to weight final keypoint selection score.

INVARIANCE PROPERTIES: Robust against textureless regions and non-repeatable dynamic environmental elements.

COMPUTER VISION BENEFIT: Eliminates false-positive visual keypoints in challenging indoor and outdoor environments.

ALGORITHM NAME: DISK (Discrete Keypoints)

PARADIGM: Reinforcement Learning Policy-Gradient Optimization

THEORETICAL FOUNDATION: Formulates keypoint detection and matching as a probabilistic decision process optimized end-to-end via policy gradient reinforcement learning (REINFORCE) using Structure-from-Motion rewards.

ALGORITHMIC STEPS:

1. Process image through CNN to generate feature maps parameterizing keypoint detection probabilities.
2. Sample discrete candidate keypoint locations using probabilistic distribution functions.
3. Extract dense descriptors and construct candidate matching graphs between image pairs.
4. Optimize detection parameters end-to-end using cycle-consistent matching accuracy rewards.

INVARIANCE PROPERTIES: Directly aligned with global multi-view 3D reconstruction quality.

COMPUTER VISION BENEFIT: Superior performance in Structure-from-Motion (SfM) and multi-view stereo pipelines.

ALGORITHM NAME: ALIKED (A Lightweight Image Keypoint Detector and Descriptor)

PARADIGM: Light-Weight Neural Architecture with Deformable Convolution

THEORETICAL FOUNDATION: Utilizes deformable convolutional blocks and dynamic feature aggregation to deliver real-time performance with high feature matching accuracy.

ALGORITHMIC STEPS:

1. Pass image through lightweight backbone augmented with deformable convolution modules.
2. Extract multi-scale feature maps using dynamic feature aggregation layers.
3. Predict keypoint detection probability maps alongside fine-grained descriptor channels.
4. Perform sub-pixel keypoint refinement using soft-argmax spatial operations.

INVARIANCE PROPERTIES: Extremely high computational speed with scale and viewpoint adaptability.

COMPUTER VISION BENEFIT: Excellent choice for resource-constrained embedded systems and real-time robotic SLAM.